

Lie Theory: Chapter 5

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1. Let V be a vector space, F be a field, and A be a subalgebra of $\mathfrak{gl}(V)$. Give an example of a function $f : A \rightarrow F$ whose eigenspace is zero.

2. Let

$$A = \left\{ \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix} : a, b \in \mathbb{C} \right\}.$$

(a) Prove that A is an abelian subalgebra of $\mathfrak{gl}(2, \mathbb{C})$.

(b) Let $\lambda : A \rightarrow \mathbb{C}$ be defined by $\lambda \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix} = a+b$. Find the weight space V_λ associated to λ .

3. (a) Let L be a Lie subalgebra of $\mathfrak{gl}(V)$. Suppose that there is a basis of V such that, with respect to this basis, every $x \in L$ is represented by a strictly upper triangular matrix. Show that L is isomorphic to a Lie subalgebra of $\mathfrak{n}(n, F)$ and hence that L is nilpotent.

(b) Prove a result analagous to (i) in the case where there is a basis of V such that with respect to this basis every $x \in L$ is represented by an upper triangular matrix.

4. Let L be a Lie algebra and let A be a subalgebra of L . The *normalizer* of A , denoted $N_L(A)$, is defined by

$$N_L(A) = \{x \in L : [x, a] \in A \text{ for all } a \in A\}.$$

Prove that $N_L(A)$ is a subalgebra of L and that $N_L(A)$ contains A . Show moreover that $N_L(A)$ is the largest subalgebra of L in which A is an ideal.

5. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $[x, y]$. Show that λ_i